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ABSTRACT

The accelerating loss of biodiversity and degradation of ecosystems due to anthropogenic activities necessitate a deeper understanding of public engagement with conservation, particularly among youth. This study investigates the knowledge, attitudes, and perceptions (KAP) of students regarding biodiversity conservation in the context of ecotourism, using Sikkim, India as a case study. A structured questionnaire was administered to 432 students across 24 educational institutions, capturing their ecological literacy, disposition towards conservation, and awareness of environmental legislation. The findings reveal a generally high level of awareness and positive attitudes towards biodiversity and ecotourism. Awareness of key pollution-control laws, particularly the Water (1974) and Air (1981) Acts, was notably low, possibly reflecting perceptions that environmental quality in the Himalayan region is relatively less threatened. Statistical analysis demonstrated acceptable to strong internal consistency across the Knowledge ($\alpha = 0.723$), Attitude ($\alpha = 0.884$), and Perception ($\alpha = 0.884$) subscales. The Kaiser–Meyer–Olkin measure (KMO = 0.891) indicated meritorious sampling adequacy for factor analysis, and Spearman's correlation analysis confirmed significant interrelationships among the KAP domains. While educational attainment significantly influenced knowledge scores, it did not uniformly affect attitudes or perceptions, suggesting the influence of cultural and contextual factors beyond formal education. The study underscores the importance of integrating experiential learning and policy literacy into environmental education to cultivate a deeper, action-oriented conservation ethic among students. Importantly, the study interprets ecotourism-related responses as indicative of conservation literacy rather than direct measures of pro-conservation behavioural orientation rather than direct measures of actual conservation behaviour. Given the cross-sectional design and the use of ecotourism used as contextual indicators, the results are associative and exploratory rather than causal. The study highlights the need for integrated environmental education, policy literacy, and experiential learning to strengthen informed conservation engagement among youth.

Keywords: Biodiversity conservation, knowledge, attitudes, perceptions, environmental legislation, educational interventions.

INTRODUCTION

Biodiversity conservation is a critical undertaking in contemporary society, especially given the alarming rate at which species are becoming extinct, and ecosystems are degrading **[1].** With environments being subject to rapid changes due to human activities, understanding public perception, particularly that of the younger generation is essential for effective conservation strategies **[2].** The youth of today will be the decision-makers of tomorrow, making it imperative that they are educated and empowered to engage in biodiversity conservation.

The bedrock of effective biodiversity conservation initiatives lies in the knowledge and awareness of the general population, particularly among students **[3].** Educational systems play a crucial role in shaping this knowledge by integrating biodiversity topics into their curriculum **[4].** Research has demonstrated that students with a solid foundation in ecological principles are more likely to appreciate the importance of biodiversity and engage in conservation efforts **[5].** Previous studies have demonstrated that students' knowledge and understanding of local biodiversity were profoundly influenced by their immediate environments, experiences, and cultural narratives **[6]**. However, knowledge alone does not necessarily lead to changes to more environmentally conscious behaviour **[7].** Students' perspectives on biodiversity conservation are shaped by myriad factors, including cultural, social, and community influences **[8].** Understanding these perspectives provides valuable insights into the collective consciousness regarding environmental stewardship among the youth **[9].** Early exposure to environmental education can assist in developing young people's perspective, preparing them to actively participate in natural resource conservation and protection as advocates for biodiversity **[10].** Further, it has been noticed that education, active involvement, and protection of natural resources change the perspectives of students towards biodiversity **[10].** Sometimes, education done poorly (preaching about "doom and gloom" scenarios) has shown to foster “why bother” attitudes in the recipients **[11].**

Therefore, it becomes imperative for emotional engagement for fostering students' perspectives towards biodiversity conservation **[12].**

Education serves as a pivotal mechanism for shaping students' attitudes towards biodiversity conservation **[4].** Research has consistently shown that well-structured educational experiences can effectively mold students’ perceptions and behaviours regarding environmental issues **[13].** Aligning educational content with students' interests and experiences significantly enhances the efficacy of biodiversity education **[14].** As indicated by studies, when educational initiatives are tailored to resonate with students’ backgrounds, they achieve better engagement results and foster positive attitudes. For instance, incorporating indigenous knowledge into biodiversity education is paramount **[4].** Previous studies have also underscored the importance of blending indigenous knowledge with scientific principles for effective resource management **[15].** By exposing students to diverse perspectives, educators can cultivate a more comprehensive understanding of biodiversity conservation, which is crucial for inspiring the youth to become proactive advocates for environmental stewardship.

This study was carried out to explore students' understanding of biodiversity and to gain insights into their knowledge, attitudes and perception (KAP) toward its conservation and protection. Additionally, the study aimed to determine whether there is a correlation between students’ understanding of biodiversity and their attitudes toward conserving it. Finally, it assessed their awareness and perceptions of existing biodiversity-related laws and policies in India.

The current study is guided by the Knowledge–Attitude–Perception (KAP) framework, which has been extensively employed in environmental education research to investigate how learners' pro-environmental orientation is collaboratively shaped by cognitive understanding, affective orientation, and perceptual awareness **[16,17].** The framework makes the assumption that acquiring knowledge can affect attitudes and perceptions, which may then lead to orientations that are ecologically conscious. Certain ecotourism-related elements were also incorporated into the current study as contextual markers of students' exposure to ideas related to conservation. However, considering the expanding global discussion over the inconsistent conservation results of ecotourism projects **[18]**, ecotourism awareness is not thought to be the same as thorough biodiversity conservation knowledge. However, considering the intricate interactions between these components, which are also context dependent, the present study adopts an exploratory rather than causal interpretation.

METHODOLOGY

Study area

The current study was conducted in the state of Sikkim (the only organic state of India) and encompasses a diverse range of educational institutions, specifically 24 schools and colleges. This region is characterized by its rich ecological and cultural landscape, creating a unique context for examining the implications of ecotourism on biodiversity conservation and community development.

The selected institutions served as vital points of engagement for understanding the perceptions, attitudes, and knowledge of local communities regarding ecotourism. The coordinates and the map of the study area are shown in **Table 1 and Figure 1.** By including a variety of schools and colleges, the study aimed to provide insights into how different educational contexts influence the understanding and adoption of ecotourism principles.

Table 1. Study area

Sl.No	Location	Latitude	Longitude
1	Nar Bahadur Bhandari Government College	27.31400	88.60060
2	Sikkim Government College, Namchi	27.17220	88.35470
3	Makha Government Senior Secondary School	27.31280	88.45560
4	Yangang Senior Secondary School	27.29560	88.41890
5	PM Shree Lingmoo Senior Secondary School	27.33440	88.46690
6	Sikkim Government College, Burtuk, Gangtok	27.34810	88.61720
7	Sikkim Government Science College, Chakung	27.14780	88.24830

8	Sanchaman Limboo Government College, Gyalshing	27.28470	88.22810
9	Chakung Senior Secondary School	27.14940	88.24220
10	Mahatma Sirijunga Senior Secondary School	27.25500	88.22940
11	Hee Yangthang Senior Secondary School	27.25500	88.19060
12	Girls Senior Secondary School, Kyongsha	27.29200	88.24530
13	Daramdin School	27.14210	88.16660
14	Tharpu School	27.14080	88.19300
15	Jorethang School	27.14670	88.30480
16	Namchi Girls School	27.17110	88.37500
17	Sikkim Alpine University	27.18350	88.34420
18	Loyola B.Ed College	27.10880	88.35000
19	Sombaria School	27.13420	88.14610
20	Soreng School	27.17140	88.20015
21	B.Ed College Soreng	27.17570	88.19840
22	Chakung School	27.15282	88.24382
23	Sribadam School	27.18305	88.26869
24	Buriakhop School	27.18305	88.26869

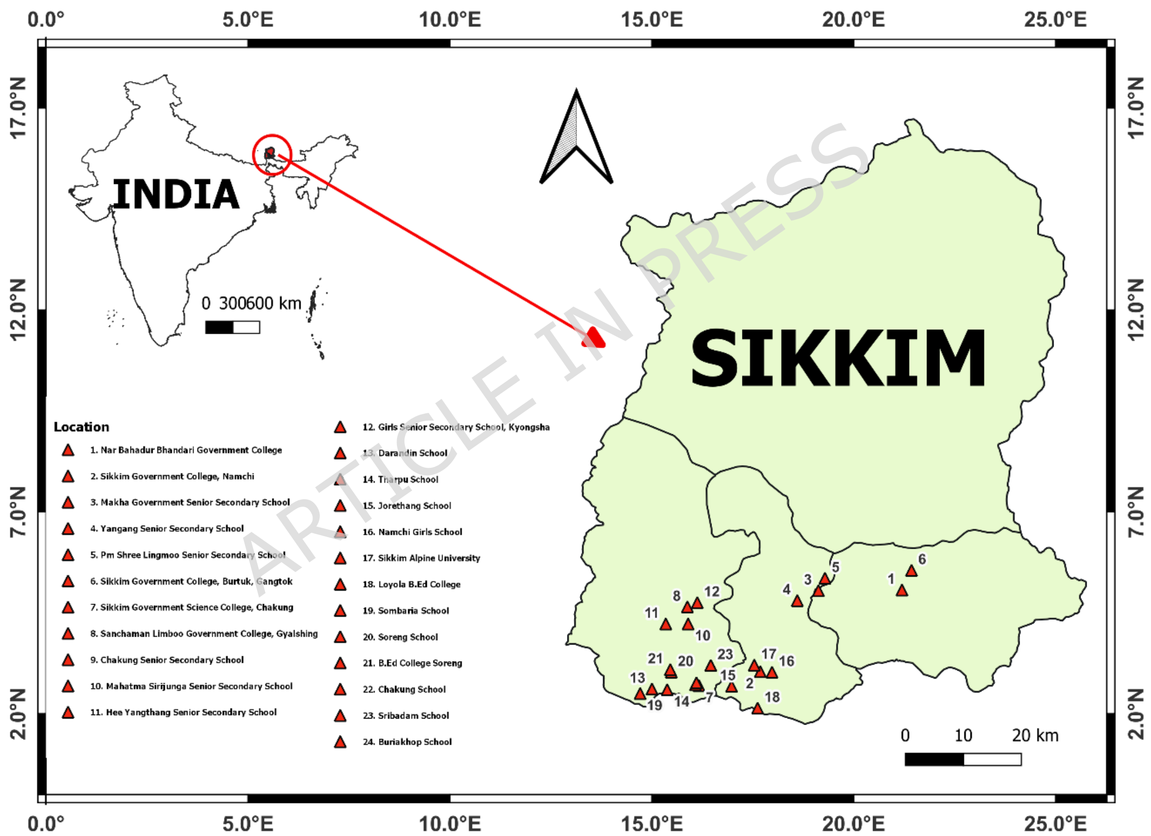

Figure 1. Map of study area

Conceptual Framework (Contextual Role of Ecotourism in the Study Framework)

The Knowledge–Attitude–Perception (KAP) continuum, which postulate that environmental knowledge may impact attitudes, which in turn influence perceptions and eventually support pro-environmental behavioural orientation, serves as the foundation for the conceptual framework of this study. The Theory of Planned Behaviour and environmental literacy models, which contend that affective disposition and behavioural intention usually follow cognitive comprehension, are in line with this sequential logic **[19-21].**

The current study treats exposure to environmental information and student demographics as background factors that could influence the development of conservation awareness. Three dimensions are used to operationalise the awareness component: perceptions of conservation and ecotourism (P), conservation attitudes (A), and biodiversity knowledge (K). The questionnaire contains items with ecotourism themes; however, they are interpreted contextually within the larger framework of biodiversity protection; the most explicit ecotourism framing is found in the perception domain. The combined influence of these domains is expected to help infer students' potential pro-conservation behavioural orientation and conservation literacy **(Figure 2).**

Ecotourism was incorporated as a contextual analytical lens rather than a direct surrogate for biodiversity conservation awareness, as previous research suggests that environmental awareness, conservation attitudes, and ecotourism perceptions are often interlinked within sustainability education frameworks **[22, 23]**. In regions such as Sikkim, where ecotourism is actively promoted through education and policy discourse, students are frequently exposed to conservation narratives framed through sustainable tourism. Therefore, ecotourism-related items were used to capture environmental interpretation, awareness, and attitudinal orientation toward conservation-linked sustainability practices.

However, the study does not assume that ecotourism awareness is equivalent to conservation behaviour. Instead, it is treated as one dimension within a broader conservation literacy construct, acknowledging that positive perceptions of ecotourism may coexist with limited ecological knowledge or behavioural engagement. The relationships among these components are often complex and context dependent; therefore, the present study adopts an exploratory rather than causal interpretation.

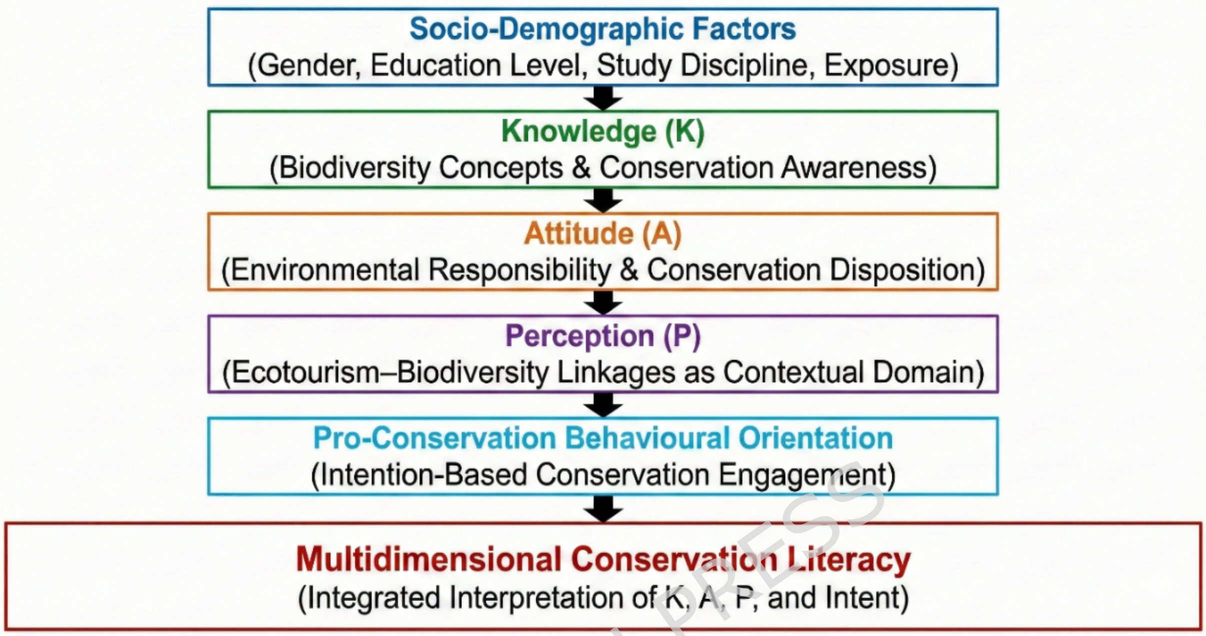

Figure 2. Conceptual framework illustrating the role of ecotourism awareness as one analytical dimension within the broader conservation literacy construct examined in this study.

Questionnaire Design

A structured questionnaire was developed to assess students' knowledge, attitudes, and perceptions regarding biodiversity conservation and ecotourism. The instrument was designed based on established environmental literacy and KAP frameworks and informed by prior studies in biodiversity conservation and ecotourism research. The questionnaire comprised five sections: (i) socio-demographic information, (ii) knowledge items (10 statements), (iii) attitude statements (10 items), (iv) perception items (10 statements), and (v) awareness of environmental legislation and policy frameworks. The knowledge, attitude, and perception (KAP) items were measured using a five-point Likert scale ranging from 1 (Strongly Disagree) to 5 (Strongly Agree), while environmental legislation awareness items were structured as multiple-response and categorical questions.

Knowledge items were constructed as objective or awareness-based questions to evaluate respondents' understanding of biodiversity conservation concepts. Attitude and perception items were measured using a five-point Likert scale ranging from strongly disagree (1) to strongly agree (5), capturing respondents' evaluative and behavioural dispositions. Such structured questionnaires are considered appropriate tools for assessment as they follow standardized measurement across respondents (Likert, 1932).

The items on the KAP questionnaire were modified to fit the local context after being taken from previously published studies **[24]**.

The questionnaire was reviewed by a panel of subject matter experts specializing in biodiversity conservation, environmental education, and ecological research. The experts evaluated each item for relevance, clarity, and contextual appropriateness for assessing students' knowledge, attitudes, and perceptions regarding biodiversity conservation and ecotourism. Their feedback was used to eliminate ambiguous or redundant items, and improve alignment with the conceptual KAP framework. The suggestions provided by the experts were incorporated into the instrument prior to the pilot survey, thereby enhancing the content validity and contextual suitability of the questionnaire.

To verify comprehensibility and reliability, a pilot test was conducted with a sample of students (n = 150). Feedback from the pilot participants helped identify minor issues related to wording and clarity. These adjustments ensured that the questionnaire was clearly understood by respondents from different educational backgrounds before the final survey was administered. Appendix I contains the completed questionnaire items and associated construct mapping.

Data collection

Data collection for the study involved the development of a questionnaire designed to gather comprehensive information on respondents' demographic characteristics, as well as their knowledge and perceptions related to biodiversity conservation. Through online platforms and convenience sampling, the surveys were distributed to students from the chosen universities and institutions, guaranteeing a wide audience and participant accessibility. Prior to final data collecting, a pilot study was carried out to improve clarity and refine unclear items. This is a recommended step in survey-based research to increase validity and reliability **[25]**.

Data analysis

The data analysis was conducted using SPSS version 26, starting with data preparation that involved importing the collected responses and performing data cleaning to remove incomplete responses and outliers. Descriptive statistics were calculated to summarize demographic characteristics, including frequency distributions, means, and standard deviations. Internal consistency reliability of the Knowledge, Attitude, and Perception (KAP) subscales was assessed using Cronbach's alpha, with a threshold value of 0.70 considered acceptable based on established psychometric guidelines **[26].** Construct validity of the KAP instrument was examined using Exploratory Factor Analysis (EFA) through Principal Component Analysis (PCA) with Direct Oblimin rotation, given the theoretical interrelatedness of the constructs. Sampling adequacy was evaluated using the Kaiser–Meyer–Olkin (KMO) measure and Bartlett's Test of Sphericity. The Kaiser criterion (eigenvalues greater than 1) was used to determine the number of components extracted. Factor loadings $\geq |0.40|$ were considered meaningful for interpretation. Normality of the data was evaluated through the Kolmogorov-Smirnov and Shapiro-Wilk tests, guiding the selection of appropriate inferential statistical methods. Mann-Whitney U test was utilised for the non-parametric data to compare differences among demographic groups. Correlation analysis using Spearman's correlation coefficient examined the relationships between knowledge, attitudes, and perceptions about biodiversity conservation. Mean ranks for Likert-scale responses were calculated to identify significant knowledge and perception areas, and results were visualized through tables, graphs, and charts to enhance comprehension of findings.

RESULTS

Normality Test

The normality of the data collected was assessed using both the Kolmogorov-Smirnov and Shapiro-Wilk tests, the results of which are documented in **Table 2.**

Table 2. Normality test

(A)	Tests of Normality					
	Kolmogorov-Smirnov[a]			**Shapiro-Wilk**		
	Statistic	**df**	**Sig.**	**Statistic**	**df**	**Sig.**
TOTAL	0.062	432	0.000	0.989	432	0.002

a. Lilliefors Significance Correction

(B)	Descriptives		**Statistic**	**Std. Error**
TOTAL	***Mean***		148.060	0.764
	95% Confidence Interval for Mean	*Lower Bound*	146.560	
		Upper Bound	149.570	
	5% Trimmed Mean		148.420	
	Median		149.000	
	Variance		252.200	
	Std. Deviation		15.881	
	Minimum		104.000	
	Maximum		190.000	
	Range		86.000	
	Interquartile Range		24.000	
	Skewness		-0.298	0.117
	Kurtosis		-0.029	0.234

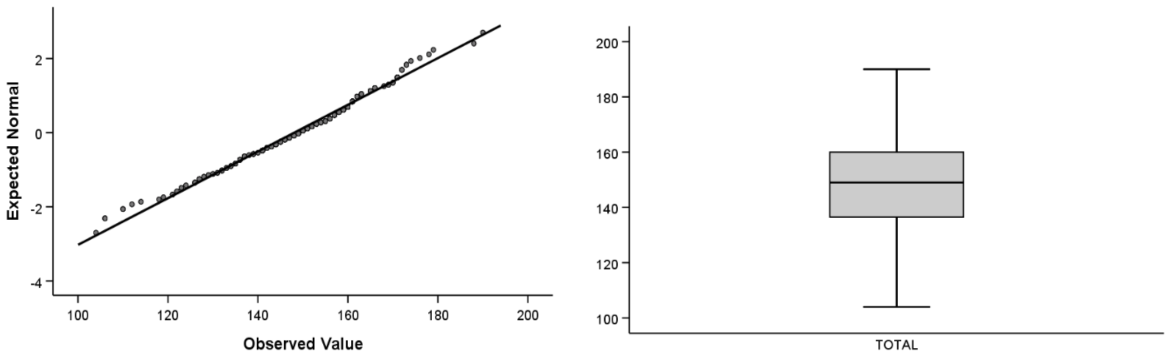

Figure 3. Normality QQ and Box plot

The Kolmogorov-Smirnov ($p < 0.001$) and Shapiro-Wilk ($p < 0.01$) tests both indicated a significant deviation from normality in the dataset. Descriptive statistics showed a mean score of 148.06 (SD = 15.88), with a 95% confidence interval ranging from 146.56 to 149.57 and a median of 149. Variance (252.20), interquartile range (24.00), skewness (-0.298), and kurtosis (-0.029) suggested a fairly symmetrical distribution with slight negative skewness. The QQ plot and box plot **(Figure 3)** further confirmed the non-normality, prompting the use of non-parametric tests for analyzing the KAP questionnaire data.

Descriptive Analysis of Demographic Variables

A total of 432 respondents participated in the study, providing a comprehensive overview of the demographic profile relevant to this research **(Table 3)**.

Table 3. Descriptive analysis of demographic variables

Characteristics		Frequency	Percent	Valid Percent	Cumulative Percent
Age	*Under 18*	144	33.33	33.33	33.33
	19-24	274	63.43	63.43	96.76
	25-34	14	3.24	3.24	100.00
	Total	**432**	**100.00**	**100.00**	
Gender	*Male*	222	51.39	51.39	51.39
	Female	210	48.61	48.61	100.00
	Total	**432**	**100.00**	**100.00**	

Education	*High school or Less*	180	41.67	41.67	41.67
	Bachelor's Degree	228	52.78	52.78	94.44
	Others (please specify)	24	5.56	5.56	100.00
	Total	**432**	**100.00**	**100.00**	

The majority of participants (63.43%) were aged 19–24, with 33.33% under 18 and only 3.24% aged 25–34, indicating a predominantly young sample. Gender distribution was nearly equal (51.39% male, 48.61% female), ensuring balanced input. In terms of education, 52.78% held a Bachelor's degree, 41.67% had high school or lower qualifications, and 5.56% fell under "Others," reflecting a reasonably literate and academically aware group.

Internal Consistency Reliability

The internal consistency of the Knowledge, Attitude, and Perception (KAP) subscales was assessed using Cronbach’s alpha **(Table 4).** The Knowledge subscale (10 items) demonstrated acceptable reliability (α = 0.723). The Attitude subscale (10 items) exhibited strong internal consistency (α = 0.884). Similarly, the Perception subscale (10 items) showed strong reliability (α = 0.884).

All corrected item–total correlations exceeded recommended minimum thresholds, indicating satisfactory homogeneity of items within each construct. These findings suggest that the instrument demonstrates acceptable to strong internal consistency across the three dimensions.

Table 4. Internal Consistency Reliability of KAP Subscales

Subscale	Number of Items	Cronbach’s Alpha
Knowledge	10	0.723
Attitude	10	0.884

Subscale	Number of Items	Cronbach's Alpha
Perception	10	0.884

Construct Validity: Exploratory Factor Analysis

Exploratory Factor Analysis (EFA) was conducted to examine the dimensional structure of the KAP instrument. Prior to extraction, sampling adequacy was evaluated using the Kaiser–Meyer–Olkin (KMO) measure and Bartlett's Test of Sphericity. The KMO value was 0.891, indicating meritorious sampling adequacy, and Bartlett's Test was significant [χ^2 (435) = 6567.684, ($p < .001$)], confirming the suitability of the data for factor analysis **(Table 5).**

Principal Component Analysis (PCA) with Direct Oblimin rotation was employed, given the theoretical interrelatedness of knowledge, attitudes, and perceptions. Using the Kaiser criterion (eigenvalues greater than 1), seven components were initially extracted, collectively explaining 63.16% of the total variance. However, inspection of the scree plot and alignment with the theoretical Knowledge–Attitude–Perception framework supported interpretation of a three-factor structure. Communalities ranged from 0.51 to 0.76, indicating adequate shared variance across items. The rotated pattern matrix is presented in **Table 6.**

Table 5. KMO and Bartlett’s Test of Sampling Adequacy

Test		Value
Kaiser-Meyer-Olkin Measure of Sampling Adequacy.		**0.891**
Bartlett’s Test of Sphericity	Approx. χ^2	**6567.684**
	Degrees of Freedom (df)	435
	Significance (p)	< 0.001

Table 6. Rotated Pattern Matrix from Principal Component Analysis (Direct Oblimin Rotation)

Item	C1	C2	C3	C4	C5	C6	C7
K1					.472		
K2					.481		
K3						.483	
K4		.615					
K5		.774					
K6				-.804			
K7		.776					
K8				-.669			
K9							.877
K10							
A1							
A2	.708						
A3				-.454			
A4	.477					-.476	
A5				-.451			
A6	.651						
A7							
A8	.770						
A9	.588						
A10	.585						
P1					.695		
P2					.777		
P3			.568				
P4	.433		.531				
P5			.507				
P6			.648				
P7							
P8			.651				
P9			.712				

Item	C1	C2	C3	C4	C5	C6	C7
P10			.410				

Mean Rank Test of the KAP Responses

Participants' responses regarding their Knowledge, Attitudes, and Perceptions (KAP) of ecotourism and biodiversity conservation were analyzed using a five-point Likert scale, where 1 indicated *"Strongly Disagree"* and 5 indicated *"Strongly Agree."* The mean rank analysis provides valuable insights into the overall awareness, sentiments, and perceptions held by the respondents, as summarized in **Tables 7A, 7B, and 7C.**

Knowledge of Ecotourism

The knowledge component, assessed across a sample of 432 participants, revealed varying degrees of understanding related to ecotourism principles and practices **(Table 7A).** The highest mean score (M = 3.85, SD = 0.88) was associated with the statement *"I understand the importance of biodiversity conservation in ecotourism,"* indicating a high level of awareness regarding the central role of biodiversity within ecotourism frameworks. Participants also demonstrated a strong understanding of the socio-economic benefits of ecotourism, with the item *"I can identify at least three benefits of ecotourism for local communities"* receiving a mean score of 3.74 (SD = 0.90). This reflects a generally good grasp of the positive local impacts of ecotourism.

In contrast, responses to regulatory knowledge were more moderate. The item *"I am aware of regulations and guidelines for responsible ecotourism practices"* received a mean of 3.42 (SD = 0.93), while awareness of endangered species in ecotourism sites was slightly higher (M = 3.45, SD = 0.86). Notably, the lowest-scoring item in the knowledge section was *"All species present in the world have been discovered already"* (M = 3.07, SD = 1.23), suggesting a prevalent misconception and highlighting the need for improved biodiversity education. Overall, the knowledge-based responses reflect a solid foundational understanding of ecotourism among participants, particularly in areas related to biodiversity and community benefits, though certain misconceptions warrant targeted educational interventions.

Attitudes Toward Ecotourism

Table 7B presents the findings related to participants' attitudes. The results suggest a generally favourable disposition toward ecotourism and its perceived contributions. The item *"I believe that ecotourism can contribute to the conservation of biodiversity"* received the highest mean score (M = 3.61, SD = 1.00), signifying strong belief in the conservation value of ecotourism. Participants also expressed support for local engagement, with statements such as *"I support the promotion of ecotourism in my community"* (M = 3.62, SD = 0.87) and *"I am willing to participate in ecotourism activities"* (M = 3.69, SD = 0.78), suggesting a readiness to contribute to ecotourism initiatives.

Perceptions of Ecotourism's Broader Implications

Participants' broader perceptions of ecotourism were examined in **Table 7C.** Among the items assessed, the perception that *"Ecotourism is a sustainable way to promote conservation"* received the highest mean score (M = 3.48, SD = 1.07), highlighting a recognition of ecotourism as a viable long-term approach to environmental stewardship. These results reflect a generally optimistic view of ecotourism's role in conservation and sustainable development, while also identifying key areas, such as regulatory awareness and biodiversity literacy where targeted interventions could enhance public understanding and engagement.

Table 7. Mean rank test of the KAP response

(A)

Code	Knowledge based questions	N	Mean	Std. Deviation	Rank
K1	I am aware of the concept of ecotourism.	432	3.58	0.86	5
K2	I can identify at least three benefits of ecotourism for local communities.	432	3.74	0.90	7
K3	I understand the importance of biodiversity conservation in ecotourism.	432	3.85	0.88	8
K4	I know the endangered species that can be found in our local ecotourism sites.	432	3.45	0.86	4
K5	I am familiar with the regulations and guidelines for responsible ecotourism practices.	432	3.42	0.93	3
K6	Biodiversity is everywhere.	432	3.88	0.92	9
K7	Biodiversity can only be measured through the species composition, specifically in the number of species and individuals.	432	3.27	0.95	2
K8	Biodiversity plays an important role in maintaining ecosystem functions (e.g., supporting, provisioning, regulating).	432	3.92	0.88	10
K9	All species present in the world have been discovered already.	432	3.07	1.23	1
K10	Changes in biological interaction like predation and parasitism can affect biodiversity.	432	3.66	0.89	6

(B)

Code	Attitude based questions	N	Mean	Std. Deviation	Rank
A1	I believe that ecotourism can contribute to the conservation of biodiversity.	432	3.61	1.00	1
A2	I support the idea of promoting ecotourism in our community.	432	3.62	0.87	2
A3	I feel positive about the role of ecotourists in raising awareness about conservation.	432	3.75	0.97	10
A4	I think ecotourism can help in generating income for local communities.	432	3.73	0.86	8
A5	I am willing to participate in ecotourism activities to support conservation efforts.	432	3.69	0.78	4
A6	I believe that ecotourism can enhance the quality of life for local residents.	432	3.63	0.89	3
A7	I value the preservation of natural habitats and wildlife in ecotourism destinations.	432	3.75	0.90	9
A8	I think ecotourism can foster a sense of pride and ownership among local residents towards conservation.	432	3.70	0.87	6
A9	I support the implementation of sustainable practices in ecotourism operations.	432	3.69	0.93	5
A10	I have a favourable attitude towards ecotourism as a tool for environmental education.	432	3.72	0.89	7

(C)

Code	Perception based questions	N	Mean	Std. Deviation	Rank
P1	I perceive ecotourism as a sustainable way to promote conservation.	432	3.48	1.07	1

P2	I believe that ecotourism can help in showcasing the unique biodiversity of our region.	432	3.69	0.93	5
P3	I perceive ecotourism as a means to empower local communities economically.	432	3.55	0.85	2
P4	I think ecotourism can create opportunities for cultural exchange and interaction.	432	3.69	0.86	4
P5	I perceive ecotourism as a way to raise awareness about environmental issues.	432	3.79	0.91	10
P6	I believe that ecotourism can enhance the reputation of our community as a conservation-friendly destination.	432	3.70	0.94	6
P7	I perceive ecotourism as a way to protect and preserve our natural heritage.	432	3.71	0.97	7
P8	I think ecotourism can foster a sense of pride and ownership among local residents towards conservation.	432	3.63	0.85	3
P9	I perceive ecotourism as a tool for promoting sustainable development in our region.	432	3.76	0.93	9
P10	I believe that ecotourism can contribute to the overall well-being of our community and environment.	432	3.75	0.95	8

Socioeconomic and Cultural Perceptions of Ecotourism

Participants showed an overall positive perception of ecotourism's socioeconomic and cultural impacts. Most recognized its role in economic empowerment (M = 3.55, SD = 0.85) and cultural exchange (M = 3.69, SD = 0.86), though awareness-raising about conservation was less emphasized (M = 3.79, SD = 0.91), indicating a need for stronger outreach.

Analysis of responses to the ten knowledge items (K1–K10) revealed that a substantial proportion of students demonstrated a moderate to high level of awareness **(Figure 4A).** On average, 48% of respondents agreed and 13% strongly agreed with the knowledge-related statements. However, a notable 27% remained neutral, indicating possible gaps in understanding or exposure to the subject. Strong disagreement was minimal, averaging around 5% or less.

The data on students' attitudes (A1–A10) reflected stronger positive alignment **(Figure 4B).** Approximately 54% of students agreed and 15% strongly agreed with statements supporting biodiversity conservation, indicating a generally favorable disposition. Neutral responses ranged from 25% to 30%, while disagreement remained low, typically under 5%.

Similarly, students' perceptions (P1–P10) were largely positive, with 50% agreement and 17% strong agreement on average **(Figure 4C).** These findings suggest that students recognize the significance of biodiversity and value its conservation. Neutral responses (~25%) suggest areas where further engagement may be needed, while negative responses were consistently low, under 10%.

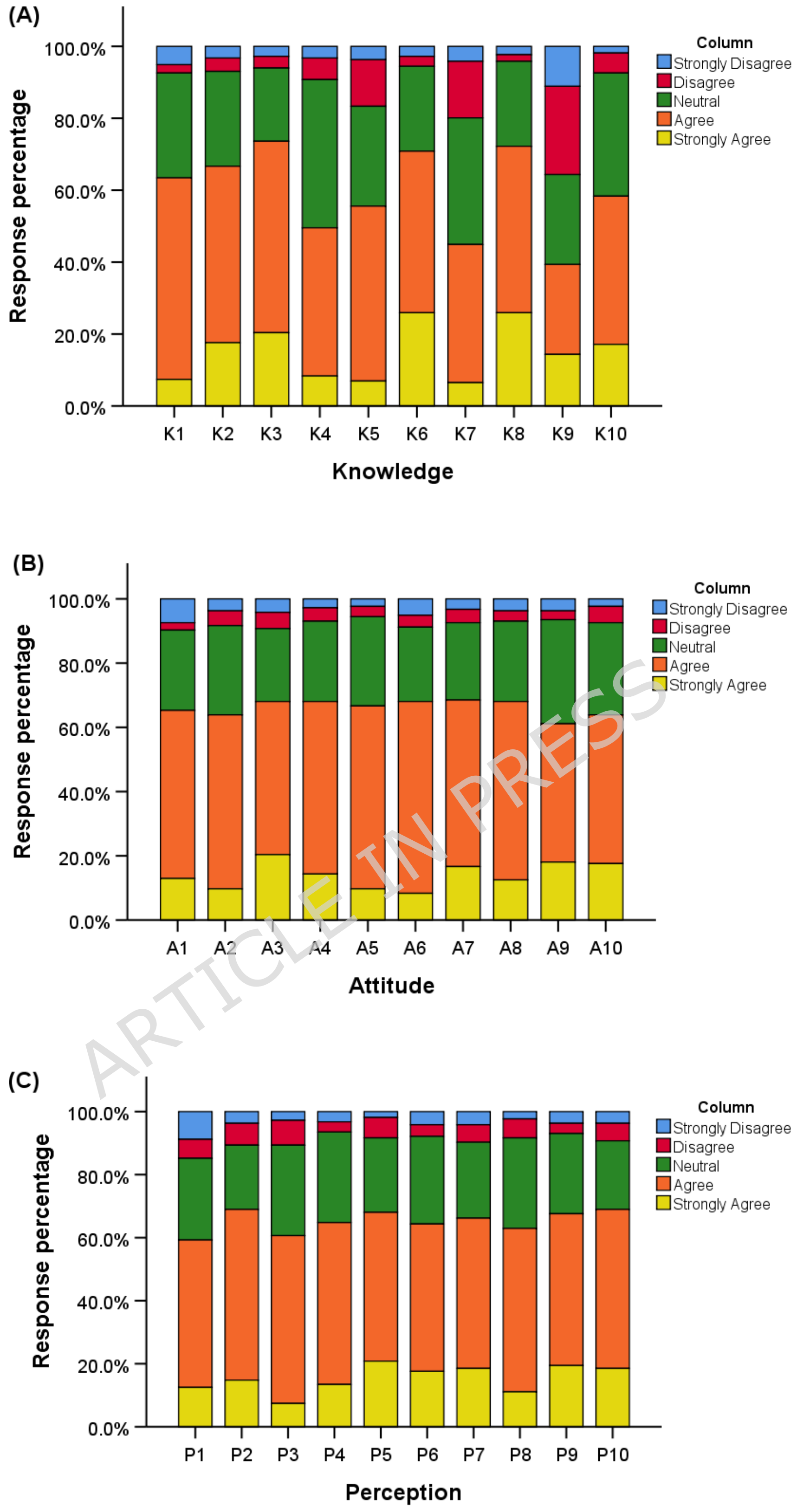

Figure 4. Response percentage of **(A)** Knowledge based questionnaire, **(B)** Attitude based questionnaire, and **(C)** Perception based questionnaire

Spearman's Correlation between KAP Levels

Spearman's correlation analysis revealed significant positive relationships among Knowledge, Attitude, and Perception (KAP) regarding ecotourism (**Table 8)**. Knowledge significantly correlated with Attitude ($\rho = 0.774$) and Perception ($\rho = 0.738$), while Attitude and Perception were also significantly correlated ($\rho = 0.792$), all at $p < 0.01$. These findings suggest that higher knowledge scores were associated with more favourable attitudes and perceptions toward ecotourism and conservation-related issues, highlighting the importance of educational initiatives to promote positive engagement and support for sustainable tourism practices.

Table 8. Spearman's correlation between KAP-level.

	Knowledge	Attitude	Perception
Knowledge	1		
Attitude	.774**	1	
Perception	.738**	.792**	1

**. Correlation is significant at the 0.01 level (2-tailed).

Response Percentage for Environmental Acts

Survey results revealed varied awareness of key environmental acts among participants **(Table 9)**. The Environment Protection Act 1986 was the most recognized (13.2%), followed by the Wildlife Protection Act 1972 (13%), and Indian Forests Act 1927 (11.9%), indicating stronger familiarity with broad conservation laws. Moderate awareness was noted for the Fisheries Act 1897 (10.1%), Forest Conservation Act 1980 (9.6%), and Biological Diversity Act 2002 (9.2%). Lower recognition was observed for the Air (prevention and control of pollution) Act 1981 and Prevention of Cruelty to Animals Act 1960 (both 8.5%), and the Water (prevention and control of pollution) Act 1974 had the least awareness (8.3%). Additionally, 7.8% cited knowledge of other acts not listed. These findings highlight uneven legal literacy and a need for improved education on environmental legislation relevant to ecotourism.

Table 9. Response percentage for Environmental Acts

Act		Responses		Percent of Cases
		N	Percent	
ACT1	Fisheries Act 1897	90	10.10%	21.10%
ACT2	Indian Forests Act 1927	106	11.90%	24.90%
ACT3	Prevention of Cruelty to Animals 1960	76	8.50%	17.80%
ACT4	Biological Diversity Act 2002	82	9.20%	19.20%
ACT5	Environment Protection Act 1986	118	13.20%	27.70%
ACT6	Air (prevention and control of pollution) Act 1981	76	8.50%	17.80%
ACT7	Forest Conservation Act 1980	86	9.60%	20.20%
ACT8	Water (prevention and control of pollution) act 1974	74	8.30%	17.40%
ACT9	Wildlife Protection Act 1972	116	13.00%	27.20%

ACT10	Other:	70	7.80%	16.40%
	Total	**894**	**100.00%**	**209.90%**

a Dichotomy group tabulated at value 1.

Response Percentage for Environmental Policies

Awareness of environmental policies varied among participants **(Table 10).** The National Biodiversity Action Plan had the highest recognition (20.8%), followed by the National Environment Policy (18%) and National Conservation Strategy (15.2%). Moderate awareness was observed for the National Agriculture Policy (12.6%), National Forest Policy and National Water Policy (both 11.2%). The National Policy on Biodiversity was recognized by 10.4% of respondents, indicating a gap in understanding biodiversity frameworks essential for ecotourism. Only 0.6% mentioned other unspecified policies, highlighting limited engagement beyond the main policy set.

Table 10. Response percentage for Environmental Policies

Policies		Responses		Percent of Cases
		N	**Percent**	
POL1	National Forest Policy	80	11.20%	19.90%
POL2	National Conservation Strategy and Policy statement on Environment and Development	108	15.20%	26.90%
POL3	National Policy and macro-level action strategy on Biodiversity	74	10.40%	18.40%
POL4	National Biodiversity Action Plan	148	20.80%	36.80%
POL5	National Agriculture Policy	90	12.60%	22.40%
POL6	National Water Policy	80	11.20%	19.90%
POL7	National Environment Policy	128	18.00%	31.80%
POL8	Other:	4	0.60%	1.00%
	Total	**712**	**100.00%**	**177.10%**

a Dichotomy group tabulated at value 1.

Role of Government and Businesses in Biodiversity conservation

The responses regarding the roles of government and businesses in biodiversity conservation were analyzed based on education levels, with a distinction made between *"Yes," "No,"* and *"Not Sure"* responses. **Figure 5** illustrates how responses were distributed among different educational groups, providing insight into public sentiment toward the responsibilities of these entities. To further understand the relationship between education level and responses, a Chi-square test was conducted **(Table 11).**

Analysis showed no significant link between education level and views on the roles of government and businesses in biodiversity conservation (Chi-square = 4.933, p = 0.294), indicating similar perceptions across educational backgrounds. However, knowledge levels varied: bachelor's degree holders demonstrated higher awareness (mean = 3.85) compared to high school graduates (mean = 3.45), highlighting the influence of education on biodiversity knowledge.

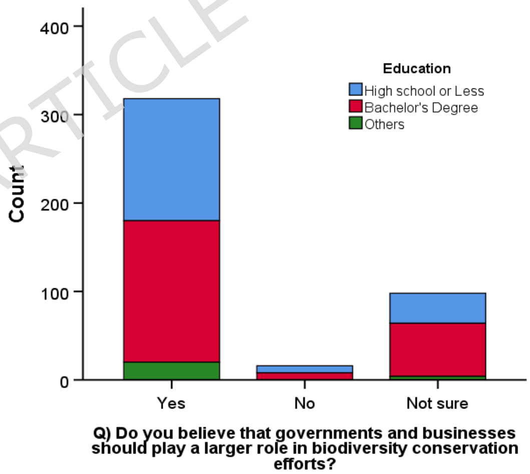

Figure 5. Response for role of Government and Businesses in Biodiversity conservation

Table 11. Role of Government and Businesses in Biodiversity conservation

	Value	df	Asymptotic Significance (2-sided)
Pearson Chi-Square	4.933[a]	4	0.294
Likelihood Ratio	5.839	4	0.211
Linear-by-Linear Association	0.703	1	0.402
N of Valid Cases	432		

a. 1 cells (11.1%) have expected count less than 5. The minimum expected count is .89.

KAP Analysis by Education Level

The analysis of Knowledge, Attitude, and Perception (KAP) concerning biodiversity conservation revealed notable distinctions between participants with high school and bachelor's degrees, as represented in **Figures 6, 7, and 8.** The responses indicated that individuals with bachelor's degrees displayed a significantly higher level of knowledge regarding biodiversity conservation compared to those with only a high school education **(Figure 6).** The mean knowledge score for bachelor's degree holders was 3.85 (SD = 0.88), while high school graduates had a mean score of 3.45 (SD = 0.86).

The analysis of attitudes towards ecotourism and biodiversity conservation showed that bachelor's degree respondents expressed more positive attitudes compared to high school participants **(Figure 7)**. Specifically, the bachelor's group reported a mean attitude score of **3.70** (SD = **0.91**), while high school graduates had a mean score of **3.61** (SD = **1.00**). This indicates that participants with a higher level of education were more inclined to agree that ecotourism contributes to biodiversity conservation and supports sustainable practices, suggesting a stronger belief in the positive impacts of ecotourism on local communities and the environment.

When examining perceptions of the roles of various stakeholders in biodiversity conservation, respondents with bachelor's degrees demonstrated a more optimistic view than their high school counterparts **(Figure 8)**. The mean perception score for bachelor's degree holders was

3.78 (SD = **0.85**), compared to **3.54** (SD = **0.92**) for high school respondents. This indicates that the bachelor's group was significantly more likely to recognize both government and businesses as critical contributors to biodiversity efforts, suggesting a nuanced understanding of their roles in implementing effective conservation strategies.

Results of the Mann-Whitney U Test

The Mann-Whitney U Test was used to compare Knowledge, Attitude, and Perception (KAP) scores on biodiversity conservation across two educational groups **(Table 12).** Among the ten knowledge items (K1–K10), significant differences were observed in K1 ($p = 0.000$), K4 ($p = 0.001$), K5 ($p = 0.020$), K7 ($p = 0.001$), and K10 ($p = 0.028$), indicating higher knowledge levels in the bachelor’s degree group for these items. The remaining knowledge items showed no significant differences.

For attitude items (A1–A10), only A4 ($p = 0.000$) showed a statistically significant difference, suggesting that bachelor's degree holders held more favorable attitudes toward that specific aspect of biodiversity conservation. No significant differences were found in the other attitude variables.

Regarding perception (P1–P10), significant differences were noted for P1 ($p = 0.023$) and P3 ($p = 0.011$), with bachelor's degree respondents showing more positive perceptions. The remaining perception items showed no significant group differences.

The findings from the hypothesis test related to various Acts and Policies concerning biodiversity conservation, conducted using the Mann-Whitney U Test, are detailed in **Table 13.**

The analysis yielded statistically significant results for ACT1 ($U = 18672.000$, $Z = -2.231$, $p = 0.026$), ACT2 ($U = 17820.000$, $Z = -3.085$, $p = 0.002$), ACT3 ($U = 18528.000$, $Z = -2.551$, $p = 0.011$), and ACT5 ($U = 17148.000$, $Z = -3.709$, $p < 0.001$), indicating significant differences between the compared groups. Therefore, the null hypotheses for these parameters were rejected.

For the remaining act-related parameters—ACT4 ($p = 0.409$), ACT6 ($p = 0.593$), ACT7 ($p = 0.859$), ACT8 ($p = 0.285$), ACT9 ($p = 0.055$), and ACT10 ($p = 0.667$)—no statistically significant differences were found, and hence, the null hypotheses for these variables were retained.

In the policy-related parameters, only POL7 showed a statistically significant result ($U = 17652.000$, $Z = -3.088$, $p = 0.002$), leading to the rejection of the null hypothesis and suggesting a significant difference in perceptions or responses regarding this policy. All other policy parameters—POL1 through POL6, POL8, and the ROLE variable—did not demonstrate statistically significant results ($p > 0.05$), and thus the null hypotheses for these parameters were retained.

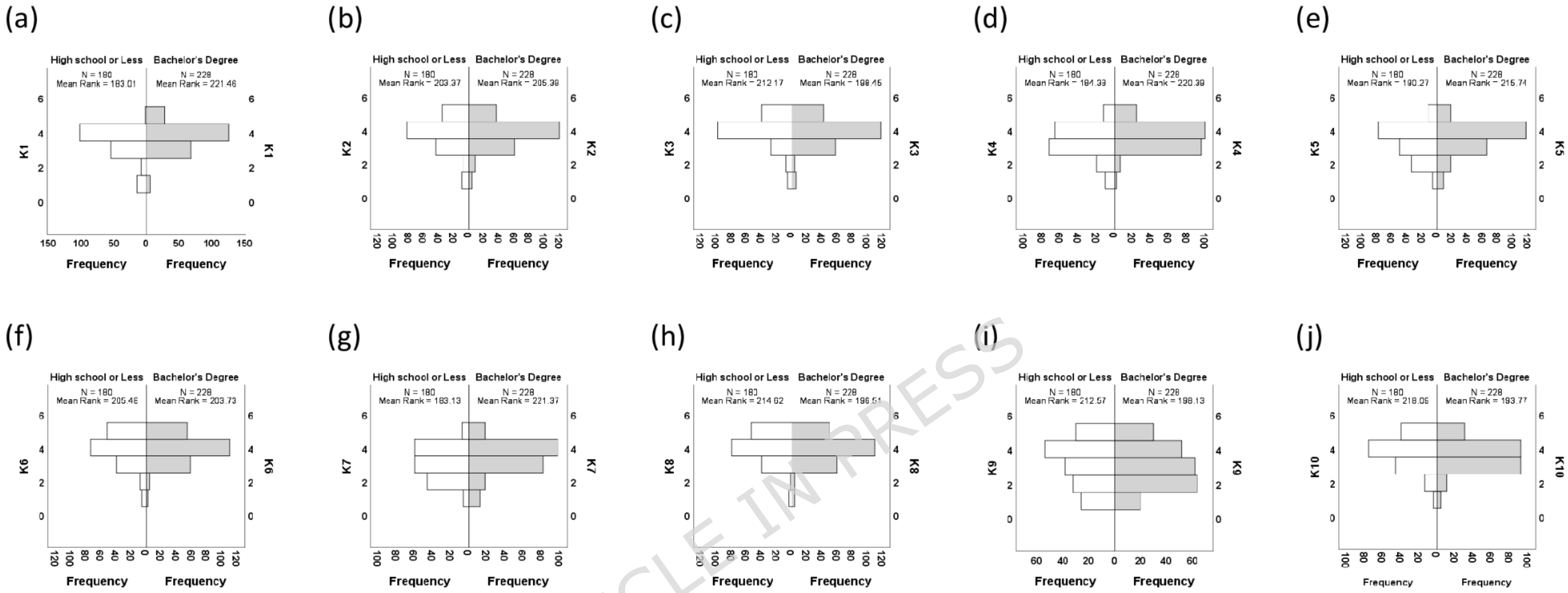

Figure 6. Response to knowledge based on education

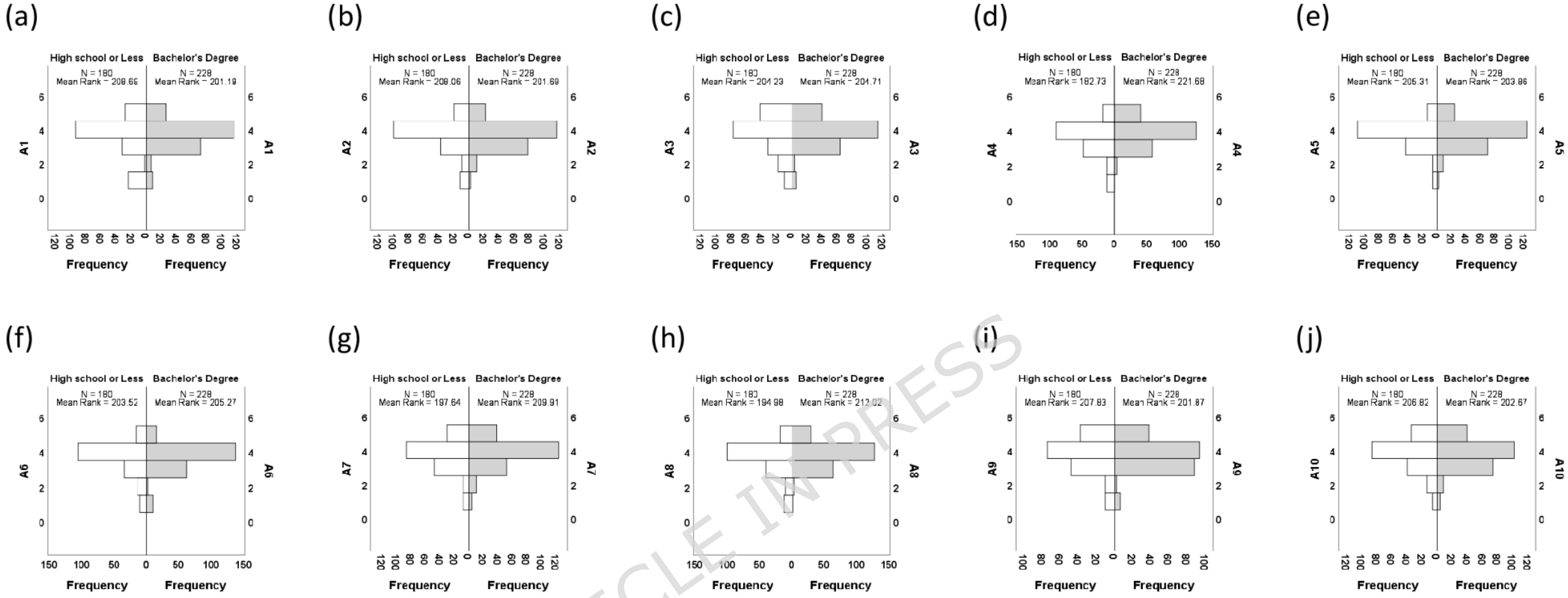

Figure 7. Response to attitude based on education

Figure 7. Response to perception based on education

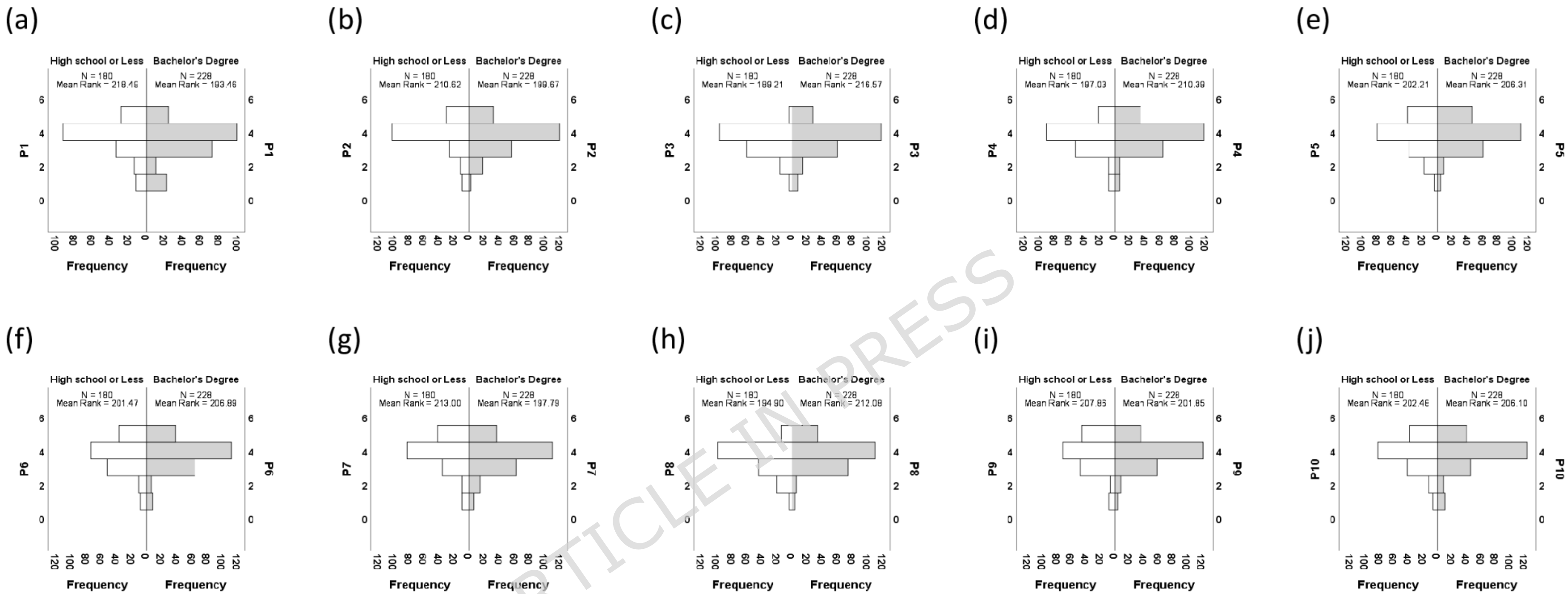

Figure 8. Response to perception based on education

Table 12. KAP hypothesis test based on Mann-Whitney U Test

(A)

Parameter	Total N	Mann-Whitney U	Wilcoxon W	Test Statistic	Standard Error	Standardized Test Statistic	Asymptotic Sig. (2-sided test)	Sig.	Decision
K1	408	24388.000	50494.000	24388.000	1056.671	3.661	0.000	0.000	Reject the null hypothesis.
K2	408	20724.000	46830.000	20724.000	1094.549	0.186	0.852	0.852	Retain the null hypothesis.
K3	408	19140.000	45246.000	19140.000	1080.142	-1.278	0.201	0.201	Retain the null hypothesis.
K4	408	24142.000	50248.000	24142.000	1097.982	3.299	0.001	0.001	Reject the null hypothesis.
K5	408	23082.000	49188.000	23082.000	1099.117	2.331	0.020	0.020	Reject the null hypothesis.
K6	408	20344.000	46450.000	20344.000	1107.488	-0.159	0.874	0.874	Retain the null hypothesis.
K7	408	24366.000	50472.000	24366.000	1119.242	3.436	0.001	0.001	Reject the null hypothesis.
K8	408	18698.000	44804.000	18698.000	1101.702	-1.654	0.098	0.098	Retain the null hypothesis.
K9	408	19068.000	45174.000	19068.000	1152.818	-1.260	0.208	0.208	Retain the null hypothesis.
K10	408	18074.000	44180.000	18074.000	1113.394	-2.197	0.028	0.028	Reject the null hypothesis.

(B)

Parameter	Total N	Mann-Whitney U	Wilcoxon W	Test Statistic	Standard Error	Standardized Test Statistic	Asymptotic Sig. (2-sided test)	Sig.	Decision
A1	408	19766.000	45872.000	19766.000	1086.719	-0.694	0.488	0.488	Retain the null hypothesis.
A2	408	19880.000	45986.000	19880.000	1075.797	-0.595	0.552	0.552	Retain the null hypothesis.
A3	408	20568.000	46674.000	20568.000	1105.954	0.043	0.965	0.965	Retain the null hypothesis.
A4	408	24438.000	50544.000	24438.000	1078.223	3.634	0.000	0.000	Reject the null hypothesis.
A5	408	20374.000	46480.000	20374.000	1050.329	-0.139	0.889	0.889	Retain the null hypothesis.
A6	408	20696.000	46802.000	20696.000	1039.450	0.169	0.866	0.866	Retain the null hypothesis.
A7	408	21754.000	47860.000	21754.000	1086.659	1.136	0.256	0.256	Retain the null hypothesis.

A8	408	22234.000	48340.000	22234.000	1062.603	1.613	0.107	0.107	Retain the null hypothesis.
A9	408	19920.000	46026.000	19920.000	1113.627	-0.539	0.590	0.590	Retain the null hypothesis.
A10	408	20102.000	46208.000	20102.000	1105.821	-0.378	0.705	0.705	Retain the null hypothesis.

(C)

Parameter	**Total N**	**Mann-Whitney U**	**Wilcoxon W**	**Test Statistic**	**Standard Error**	**Standardized Test Statistic**	**Asymptotic Sig. (2-sided test)**	**Sig.**	**Decision**
P1	408	18002.000	44108.000	18002.000	1106.570	-2.276	0.023	0.023	Reject the null hypothesis.
P2	408	19418.000	45524.000	19418.000	1075.452	-1.025	0.306	0.306	Retain the null hypothesis.
P3	408	23272.000	49378.000	23272.000	1077.075	2.555	0.011	0.011	Reject the null hypothesis.
P4	408	21864.000	47970.000	21864.000	1085.161	1.239	0.216	0.216	Retain the null hypothesis.
P5	408	20932.000	47038.000	20932.000	1106.647	0.372	0.710	0.710	Retain the null hypothesis.
P6	408	21066.000	47172.000	21066.000	1107.067	0.493	0.622	0.622	Retain the null hypothesis.
P7	408	18990.000	45096.000	18990.000	1106.114	-1.383	0.167	0.167	Retain the null hypothesis.
P8	408	22248.000	48354.000	22248.000	1084.879	1.593	0.111	0.111	Retain the null hypothesis.
P9	408	19916.000	46022.000	19916.000	1101.107	-0.549	0.583	0.583	Retain the null hypothesis.
P10	408	20884.000	46990.000	20884.000	1093.018	0.333	0.739	0.739	Retain the null hypothesis.

Table 13. Acts, Policies and Role hypothesis test based on Mann-Whitney U Test

(A)

Parameter	**Total N**	**Mann-Whitney U**	**Wilcoxon W**	**Test Statistic**	**Standard Error**	**Standardized Test Statistic**	**Asymptotic Sig. (2-sided test)**	**Sig.**	**Decision**
ACT1	408	18672.000	44778.000	18672.000	828.302	-2.231	0.026	0.026	Reject the null hypothesis.
ACT2	408	17820.000	43926.000	17820.000	875.126	-3.085	0.002	0.002	Reject the null hypothesis.
ACT3	408	18528.000	44634.000	18528.000	780.930	-2.551	0.011	0.011	Reject the null hypothesis.
ACT4	408	19848.000	45954.000	19848.000	813.314	-0.826	0.409	0.409	Retain the null hypothesis.

ACT5	408	17148.000	43254.000	17148.000	909.036	-3.709	0.000	0.000	Reject the null hypothesis.
ACT6	408	20928.000	47034.000	20928.000	763.432	0.534	0.593	0.593	Retain the null hypothesis.
ACT7	408	20664.000	46770.000	20664.000	813.314	0.177	0.859	0.859	Retain the null hypothesis.
ACT8	408	19704.000	45810.000	19704.000	763.432	-1.069	0.285	0.285	Retain the null hypothesis.
ACT9	408	22272.000	48378.000	22272.000	914.179	1.916	0.055	0.055	Retain the null hypothesis.
ACT1O	408	20832.000	46938.000	20832.000	725.507	0.430	0.667	0.667	Retain the null hypothesis.

(B)

Parameter	**Total N**	**Mann-Whitney U**	**Wilcoxon W**	**Test Statistic**	**Standard Error**	**Standardized Test Statistic**	**Asymptotic Sig. (2-sided test)**	**Sig.**	**Decision**
POL1	408	19212.000	45318.000	19212.000	805.528	-1.624	0.104	0.104	Retain the null hypothesis.
POL2	408	20040.000	46146.000	20040.000	868.939	-0.552	0.581	0.581	Retain the null hypothesis.
POL3	408	20748.000	46854.000	20748.000	772.296	0.295	0.768	0.768	Retain the null hypothesis.
POL4	408	20340.000	46446.000	20340.000	969.165	-0.186	0.853	0.853	Retain the null hypothesis.
POL5	408	19716.000	45822.000	19716.000	835.514	-0.962	0.336	0.336	Retain the null hypothesis.
POL6	408	20568.000	46674.000	20568.000	780.930	0.061	0.951	0.951	Retain the null hypothesis.
POL7	408	17652.000	43758.000	17652.000	928.788	-3.088	0.002	0.002	Reject the null hypothesis.
POL8	408	20568.000	46674.000	20568.000	201.835	0.238	0.812	0.812	Retain the null hypothesis.
ROLE	408	21956.000	48062.000	21956.000	914.641	1.570	0.116	0.116	Retain the null hypothesis.

DISCUSSION

Demographical Insights and Participant Profile

Demographically, the study sample (N = 432) was predominantly composed of youth, with 63.43% aged 19–24 and 33.33% under 18. This skew toward a younger cohort is particularly salient, as it frames the study's findings within the context of emerging generational attitudes towards environmental issues. Youth engagement is a critical vector in the sustainability discourse, and the demographic profile herein underscores both the urgency and opportunity for targeted interventions among this age group. The near-equal gender representation (51.39% male; 48.61% female) enhances the generalizability of the findings by mitigating gender-based response biases. Prior studies have demonstrated that understanding of biodiversity varies by gender **[27].** Furthermore, the educational distribution with over half (52.78%) holding a Bachelor's degree, suggests a sample with sufficient cognitive and academic capacity to engage meaningfully with the constructs under investigation.

Knowledge, Attitude and Perception toward Ecotourism

Knowledge Domain

The descriptive analysis of the KAP domains reveals a generally encouraging level of conservation awareness among participants, with ecotourism-related responses reflecting supportive orientations within this broader environmental context, though certain knowledge and perceptual aspects remain underdeveloped. Specifically, participants demonstrated substantial understanding of biodiversity conservation and the socio-economic benefits of ecotourism, as reflected in high mean scores for items assessing foundational knowledge and community impacts. Sikkim has actively promoted community-based ecotourism, involving residents in tourism activities such as homestays, guides, hotel operators, and taxi drivers, which may have contributed to greater student awareness of ecotourism benefits. Besides, tourism has incentivised the maintenance and showcasing of traditional customs, festivals, and crafts, reinforcing cultural identity **[28].**

However, responses to regulatory knowledge were more moderate, and misconceptions regarding biodiversity discovery (e.g., *"All species have already been discovered"*) highlight significant educational gaps. Addressing this misconception may require more experiential and inquiry-based teaching strategies in environmental education. For instance, classroom activities such as local biodiversity documentation, species inventory projects, or participation in citizen science initiatives (e.g., biodiversity monitoring programs) can help students recognize that scientific discovery is an ongoing process. Such experiential learning approaches allow students to directly observe biodiversity in their surroundings and understand that many species remain undocumented, thereby reinforcing the dynamic and evolving nature of biodiversity research. Further, only 40% of the participants were well-versed with regulations and guidelines related to ecotourism. These results are consistent with earlier studies showing that although environmental consciousness is increasing generally, knowledge of governance and scientific biodiversity concepts is still lacking **[29,30].** Studies have shown that even among professional tourist guides, there can be gaps in understanding ecotourism principles, indicating a need for more comprehensive education and training programs **[31].**

Attitude Domain

Participants' attitudes toward ecotourism were notably positive, with a majority expressing strong support for its conservation value and community engagement potential. The high levels of willingness to participate in ecotourism initiatives are particularly promising for participatory conservation models. This resonates with earlier studies suggesting that favourable attitudes are key predictors of behavioural intention and sustainable engagement in conservation programs **[19, 32].**

Perception Domain

The perception domain further affirmed this optimism, though results suggested slightly more ambivalence in recognizing the broader social and cultural implications of ecotourism. The fact that ecotourism strengthens cultural identification is noteworthy **[28],** many respondents perceived ecotourism as a tool for sustainable development and intercultural exchange. However, fewer youths recognized its role in fostering local pride or environmental stewardship. Previous studies have found that local residents of Laguna Province, Philippines recognised the benefits of ecotourism but were less aware of how it may foster a sense of environmental responsibility and pride in regional history **[33]**. Crucially, the Spearman's correlation analysis revealed strong and statistically significant associations among all three KAP domains. The robust correlation between Knowledge and Attitude ($\rho = 0.774$, $p < 0.01$) is consistent with rationalist models suggesting that environmental education is associated with more favourable environmental attitudes towards the environment **[34].** Likewise, the positive association between Knowledge and Perception ($\rho = 0.738$, $p < 0.01$) highlights the significance of knowledgeable comprehension in influencing how people see the wider ramifications of ecotourism.

Interrelationship among KAP Domains

The substantial positive association between attitude and perception ($\rho = 0.792$, $p < 0.01$) is consistent with well-known behavioural models like the Theory of Planned Behaviour **[19]**, which holds that attitudes have a major impact on behavioural intentions and perceptions. The fact that a visitor's attitude towards a place is impacted by how they perceive the place's image recognises the significance of having a good attitude in controlling perception **[32, 35].** Significant correlations between the three KAP dimensions support earlier findings indicating students who knew more about the environment had more positive attitudes and actions related to the environment **[36].**

Environmental Legal Literacy

Regarding environmental legal literacy, the varied levels of recognition across legislative frameworks indicate an uneven understanding of environmental governance among respondents. Media exposure, curriculum prioritisation, and active promotion through government and non-governmental organisation campaigns are all responsible for the disproportionate awareness of some environmental legislation, such as the Wildlife Protection Act of 1972 and the Environment Protection Act of 1986. These laws have been at the core of high-profile environmental crises (like the Bhopal Disaster) and conservation initiatives (like Project Tiger), which have increased their prominence in the media and in discussions about policy **[37, 38].**

Conversely, alarmingly low awareness of other essential environmental laws, such as the Water Act, 1974, and the Air Act, 1981 (8.3% and 8.5%), points to critical gaps in legal literacy that could undermine ecotourism goals. According to the Global Education Monitoring team's 2016 report, those who have gained more education place a higher priority on protecting the environment and solving environmental issues **[39].** With an emphasis on how humans affect the environment, environmental education has been incorporated into school curricula worldwide in recent years **[39],** perhaps raising awareness of water issues among younger students. The alarmingly low awareness of other essential environmental laws, such as the Water Act, 1974, and the Air Act, 1981 (8.3% and 8.5%) points to critical gaps in legal literacy that could undermine ecotourism goals. One explanation for the participants' limited awareness may be media reports of Northeast India's superior air quality **[40],** when compared to other regions of the nation. However, it is important to remember that the Himalayan air quality has significantly worsened in recent years, surpassing the World Health Organization's (WHO) yearly recommendations **[41].** Northeast India and the Himalayan ranges are vulnerable to temperature fluctuations, precipitation variations, ice melting, and changes in hydrological cycles due to air pollution, which causes ecological disruptions and global warming **[42].**

Awareness of biodiversity-specific laws like the Biological Diversity Act, 2002 (9.2%) and the Forest Conservation Act, 1980 (9.6%) also remains limited, reflecting a persistent gap in understanding frameworks that are central to ecological stewardship. These results underscore the need for targeted, context-specific legal education programs, particularly in regions where ecotourism is being developed or promoted. The solution of environmental problems directly depends on the level of environmental literacy and public education **[43-45].** Previous studies have reported that high school pupils and University students note a low role of Law in the development of environmental and legal literacy **[46].** Many of them believe that they have never considered environmental safety and protection from a legal point of view.

Nonetheless, the relationship between legal knowledge and ecotourism comprehension points to a potential area of cooperation: presenting laws as instruments for environmental justice and intergenerational equity rather than merely as regulatory ones, which could ultimately safeguard the local community and resources while preserving sustainability **[47, 48].** Furthermore, the large number of visitors is displacing natural woods with concrete jungles **[38],** which may have long-term negative effects on the ecosystem and call into question the viability of ecotourism.

Public Awareness of Environmental Policies

An interesting outcome of this study was that participants demonstrated relatively higher awareness of key environmental policy frameworks such as the National Biodiversity Action Plan (NBAP), the National Conservation Strategy and Policy Statement on Environment and Development, and the National Environment Policy (NEP). This trend may be attributed to a combination of curricular exposure and the active role of state institutions in environmental education. Sikkim, being a biodiversity-rich Himalayan state, has historically emphasized environmental stewardship through its Green State mission, organic farming initiatives, and integration of conservation themes in school curricula, possibly contributing to this awareness **[50, 51].** Furthermore, the presence of various awareness campaigns, eco-club activities **[52],** and NGO-led biodiversity programs in schools and colleges could have facilitated early engagement with these national frameworks. This finding suggests that state-level environmental governance, when coupled with institutional and educational support, can enhance awareness of national environmental agendas among young learners.

However, the National Policy and Macro-level Action Strategy on Biodiversity was recognized by only 10.4% of participants, highlighting a potential gap in awareness of comprehensive strategies for biodiversity management.

Perceptions of Governmental and Corporate Roles in Biodiversity Conservation

There was no discernible correlation between responses and educational level in the stratified analysis of opinions about the roles of corporations and the government in biodiversity conservation. This implies that people from different educational backgrounds have a common awareness of the roles that businesses and governments play in conservation initiatives.

The general public's expectations for institutional accountability in environmental stewardship may be reflected in this consistency in perception. Corporate responsibility in environmental conservation is promoted by the National Voluntary Guidelines on Social, Environmental, and Economic Responsibilities of Business (NVGs), which were first introduced by the Ministry of Corporate Affairs in 2011. The NVGs' guiding principles, which emphasise the value of teamwork in biodiversity conservation, are consistent with the seeming popular consensus regarding the roles of corporations and the government.

Knowledge, Attitude, and Perception (KAP) Analysis

The KAP analysis revealed that individuals with bachelor's degrees exhibited significantly higher knowledge scores regarding biodiversity conservation compared to those with only a high school education (mean scores: 3.85 vs. 3.45, respectively). This finding aligns with existing literature that correlates higher educational attainment with increased environmental awareness and pro-environmental behaviours **[53].**

However, the absence of significant differences in attitudes and perceptions across educational levels suggests that factors beyond formal education, such as cultural values, community engagement, and exposure to environmental initiatives, may influence public attitudes toward biodiversity conservation. This highlights the importance of inclusive educational programs and community-based conservation initiatives that transcend educational disparities **[34].** Previous studies have referred to this phenomenon as the “value action-gap” where they argue that knowledge and attitude alone are insufficient predictors of behaviour because an individual's actions are affected by many internal and external factors **[34].** This perceptual gap could hinder long-term community investment in conservation and often results in community apathy, conflict, and failure of tourism ventures **[54].** The lower acknowledgment of community pride and ownership may reduce tourist satisfaction thereby impacting both visitor perception and economic returns **[55].**

Perceptions of Stakeholder Roles

Perception scores also differed notably by educational attainment, with bachelor's degree holders displaying a higher tendency to acknowledge government and corporate actors as key players in biodiversity conservation (mean: 3.78 vs. 3.54). This supports earlier findings that higher education facilitates more nuanced understandings of environmental governance structures **[56].**

Implications for Policy and Environmental Education

These findings collectively underscore the critical role of education in shaping both cognitive (knowledge) and affective (attitude/perception) components of public engagement with biodiversity policy. While general awareness of biodiversity legislation exists across educational levels, targeted educational outreach is necessary to deepen understanding of specific policies and to foster the critical thinking needed for effective civic participation in conservation initiatives **[51].** Moreover, aligning environmental education with ecotourism narratives may serve as a dual strategy to enhance public awareness while supporting sustainable local development.

CONCLUSION

This study examined the knowledge, attitudes, and perceptions (KAP) of students in Sikkim regarding biodiversity conservation within an ecotourism context using a structured questionnaire and non-parametric analysis. The results show a supportive cognitive and affective orientation toward conservation issues, with moderate to high biodiversity awareness, generally positive attitudes, and good perceptions toward conservation and ecotourism-related sustainability.

Due to the cross-sectional and self-reported nature of the data, significant positive connections between knowledge, attitudes, and perceptions imply interrelated aspects of conservation literacy; however, these relationships are associative rather than causative. Higher knowledge scores were linked to educational attainment, but attitudes and perceptions varied little between educational groups, suggesting the impact of more extensive contextual and experiential factors.

There are still noticeable gaps in regulatory and environmental policy literacy, despite encouraging approaches. Items pertaining to ecotourism served as contextual markers of conservation awareness, and under the KAP framework, a willingness to engage in ecotourism activities should be regarded as an intention and an attitude rather than as actual behaviour. All things considered, the study shows a rather high but unequal profile of conservation literacy and emphasises the necessity of improving young biodiversity education, policy literacy, and context-based environmental learning.

Limitations of the study

The current study has a number of limitations that should be noted, even if it offers insightful information about students' awareness of conservation. First, causal inference is not possible since the cross-sectional design only records relationships between the knowledge, attitudes, and sensory domains at one particular moment in time.

Therefore, rather than being deterministic, the directional linkages suggested in the conceptual framework should be viewed as exploratory.

Second, the study uses self-reported responses, which could be impacted by respondents' subjective interpretations of survey items or social desirability bias. Perceptual measures may still entail individual response variability even when the instrument was modified from well-established research and reliability was evaluated.

Third, ecotourism awareness does not fully represent overall biodiversity conservation competence, even if ecotourism-themed items were included to give respondents in Sikkim a locally relevant environmental context. Therefore, rather than being conclusive markers of conservation behaviour, the results should be evaluated within the larger framework of conservation literacy.

Lastly, rather than observing pro-conservation behaviours, the study mainly evaluated behavioural orientation and purpose. Future studies using mixed-method techniques, longitudinal designs, or behavioural observations would support and expand on the current findings. The absence of qualitative insights also limits the depth of interpretation regarding how respondents conceptualize “ecotourism” in relation to “biodiversity conservation.” As the inclusion of qualitative components was not feasible within the scope of the present survey design, future research may incorporate open-ended responses, interviews, or mixed-method approaches to better capture nuanced interpretations and contextual meanings associated with ecotourism and conservation awareness. Such approaches would further strengthen the interpretative validity and contextual grounding of conservation literacy assessments.

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DECLARATIONS

Ethics approval and consent to participate

This research involved collecting anonymous questionnaire responses from students regarding biodiversity conservation. According to the Sikkim Alpine University Ethics Committee (EC) policy, this minimal-risk study using anonymous, non-sensitive data was waived by the Ethics Committee. Ethical procedures followed the principles of the Declaration of Helsinki (1964, as amended).

Informed Consent

Participants were informed about the purpose of the study, the voluntary nature of participation, and the confidentiality of their responses before completing the questionnaire. Participation was entirely voluntary, and respondents were free to decline participation or withdraw at any time. The questionnaire was administered anonymously, and all responses were used solely for academic research purposes and reported in aggregated form to ensure confidentiality. Informed consent was obtained from all participants, and for participants under the age of 16 years, informed consent was obtained from their parents or legal guardians.

Consent for publication

Not applicable.

Availability of data and material

The datasets generated and/or analyzed during the current study are available from the corresponding author upon reasonable request.

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Conflict of Interest

The authors declare that they have no conflict of interest.

Clinical Trial

Clinical trial number: not applicable.

Authors' contributions

Leela Sharma, Sheela Gurung, Meenu Chettri, Monu Hangma Limbu assisted in the review of literature, designing the methodology and collected data. Tabbasum Banu and Shristie Subba assisted in data collection, review of literature, and data analysis. Dr. Subhankar Gurung and Dr. Aditya Moktan Tamang supervised the project, performed the final data interpretation, and wrote the manuscript. All authors read and approved the final manuscript.

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APPENDIX I

Table 1: Construct Mapping of Questionnaire Items to Knowledge (K), Attitude (A), and Perception (P) Domains with Measurement Type and Analytical Role in the KAP Framework of Biodiversity Conservation and Ecotourism Study

Construct	**Domain Description**	**Item Codes***	**Measurement Type**	**Role in Analytical Model**
Knowledge (K)	Cognitive understanding of biodiversity concepts and conservation awareness	K1–K10	Structured awareness items (ordinal/score-based)	Predictor Variable
Attitude (A)	Evaluative disposition toward biodiversity conservation and environmental responsibility	A1–A10	5-point Likert scale (ordinal)	Affective Mediator
Perception (P)	Awareness and interpretation of ecotourism–biodiversity linkages	P1–P10	5-point Likert scale (ordinal)	Contextual Perceptual Domain

Table 2: Structured Questionnaire for Assessing Knowledge, Attitude, and Perception (KAP) toward Biodiversity Conservation and Ecotourism among Respondents

Section A: Demographic Information

1. Age
   - Under 18
   - 19–24
   - 25–34
   - 35 or older
2. Gender
   - Male
   - Female
   - Prefer not to say
3. Education Level
   - High School or Less
   - Bachelor’s Degree
   - Other (Specify)

Section B: Knowledge (K)

(5-point Likert scale: Strongly Disagree – Strongly Agree)

K1. I am aware of the concept of ecotourism.

K2. I can identify at least three benefits of ecotourism for local communities.

K3. I understand the importance of biodiversity conservation in ecotourism.

K4. I know the endangered species that can be found in our local ecotourism sites.

K5. I am familiar with the regulations and guidelines for responsible ecotourism practices.

K6. Biodiversity is everywhere.

K7. Biodiversity can only be measured through species composition (number of species and individuals).

K8. Biodiversity plays an important role in maintaining ecosystem functions (supporting, provisioning, regulating).

K9. All species present in the world have already been discovered.

K10. Changes in biological interactions (e.g., predation and parasitism) can affect biodiversity.

Section C: Attitude (A)

(5-point Likert scale: Strongly Disagree – Strongly Agree)

A1. I believe that ecotourism can contribute to the conservation of biodiversity.
A2. I support the idea of promoting ecotourism in our community.
A3. I feel positive about the role of ecotourists in raising awareness about conservation.
A4. I think ecotourism can help generate income for local communities.
A5. I am willing to participate in ecotourism activities to support conservation efforts.
A6. I believe that ecotourism can enhance the quality of life for local residents.
A7. I value the preservation of natural habitats and wildlife in ecotourism destinations.
A8. I think ecotourism can promote cultural exchange and understanding.
A9. I support the implementation of sustainable practices in ecotourism operations.
A10. I have a favorable attitude toward ecotourism as a tool for environmental education.

Section D: Perception (P)

(5-point Likert scale: Strongly Disagree – Strongly Agree)

P1. I perceive ecotourism as a sustainable way to promote conservation.
P2. I believe that ecotourism can help showcase the unique biodiversity of our region.
P3. I perceive ecotourism as a means to empower local communities economically.
P4. I think ecotourism can create opportunities for cultural exchange and interaction.
P5. I perceive ecotourism as a way to raise awareness about environmental issues.
P6. I believe that ecotourism can enhance the reputation of our community as a

conservation-friendly destination.

P7. I perceive ecotourism as a way to protect and preserve our natural heritage.

P8. I think ecotourism can foster a sense of pride and ownership among local residents toward conservation.

P9. I perceive ecotourism as a tool for promoting sustainable development in our region.

P10. I believe that ecotourism can contribute to the overall well-being of our community and environment.

Section E: Awareness of Environmental Laws and Policies

(Select all that apply)

Indian Environmental Laws:

- Fisheries Act (1897)
- Indian Forest Act (1927)
- Prevention of Cruelty to Animals Act (1960)
- Wildlife Protection Act (1972)
- Water (Prevention and Control of Pollution) Act (1974)
- Forest Conservation Act (1980)
- Air (Prevention and Control of Pollution) Act (1981)
- Environment Protection Act (1986)
- Biological Diversity Act (2002)

National Policies:

- National Forest Policy
- National Biodiversity Action Plan
- National Environment Policy
- National Water Policy
- National Conservation Strategy and Policy Statement on Environment and Development
- National Policy and Macro-Level Action Strategy on Biodiversity
- National Agriculture Policy

Additional Questions:

1. Do you believe that governments and businesses should play a larger role in biodiversity conservation efforts?
    - Yes

- No
- Not Sure